ORGNZ-07: Advanced Permission Patterns

> **Series:** ORGNZ — Organize Data: Buckets, Segments, Security | **Notebook:** 7 of 10 | **Created:** January 2026 | **Last Updated:** 05/06/2026

Overview

This notebook covers advanced permission patterns including record-level permissions, field-based access, and combining multiple access control mechanisms for enterprise-scale data governance.

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS environment with Grail enabled
Permissions	Account admin or IAM policy management access
Knowledge	Completed ORGNZ-06 (Security Context)
Data	At least 1 hour of log data

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Learning Objectives

By the end of this notebook, you will:
- Implement record-level permissions using various attributes
- Understand field-level access restrictions
- Use policy boundaries for reusable condition management
- Combine bucket, record, and field-level policies
- Design enterprise-scale permission architectures

Record-Level Permissions

Record-level permissions filter data at query time based on record attributes:

Supported Record-Level Conditions

Condition	Description	Example
`storage:k8s.namespace.name`	Kubernetes namespace	`= 'production'`
`storage:k8s.cluster.name`	Kubernetes cluster	`= 'main-cluster'`
`storage:host.name`	Host name	`= 'web-server-01'`
`storage:dt.host_group.id`	Host group	`STARTSWITH 'prod-'`
`storage:aws.account.id`	AWS account	`= '123456789012'`
`storage:gcp.project.id`	GCP project	`= 'my-project'`
`storage:azure.subscription`	Azure subscription	`= 'sub-id'`
`storage:azure.resource.group`	Azure resource group	`= 'my-rg'`
`storage:dt.security_context`	Custom context	`MATCH ('team-*')`

Record-Level Policy Examples

Kubernetes Namespace Isolation

```
{
  "name": "production-namespace-access",
  "description": "Access only to production namespace data",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read WHERE
storage:k8s.namespace.name = 'production';"
}
```

Host Group Based Access

```
{
  "name": "web-tier-access",
  "description": "Access to web tier hosts only",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read, storage:metrics:read WHERE
storage:dt.host_group.id STARTSWITH 'web-';"
}
```

Combined Namespace and Host Group

```
{
  "name": "prod-web-tier-access",
  "description": "Production web tier only",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read WHERE
storage:k8s.namespace.name = 'production' AND storage:dt.host_group.id
STARTSWITH 'web-';"
}
```

Cloud Account Isolation

```
{
  "name": "aws-team-account-access",
  "description": "Access to specific AWS account",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read, storage:metrics:read,
storage:spans:read WHERE storage:aws.account.id = '123456789012';"
}
```

Field-Level Access

Control access to specific fields within records:

Use Case	Implementation
Hide sensitive fields	Deny access to specific fields
Mask PII	Field-level policies

Use Case	Implementation
Compliance requirements	Restrict access to audit fields

Field-Level Policy Example

```
{
  "name": "restricted-field-access",
  "description": "Hide sensitive fields from general users",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read; DENY
storage:logs:read:user.email, storage:logs:read:user.ip_address;"
}
```

Combined Permission Patterns

Pattern 1: Layered Access Control

Combine bucket and record-level permissions:

```
Layer 1: Bucket access
  ALLOW storage:buckets:read WHERE bucket-name IN ('team_logs',
'shared_logs')

Layer 2: Record filtering
  ALLOW storage:logs:read WHERE k8s.namespace.name = 'team-namespace'

Layer 3: Field masking (optional)
  DENY storage:logs:read:sensitive_field
```

Pattern 2: Multi-Team Shared Bucket

Multiple teams share a bucket with security context isolation:

```
// Team A policy
{
  "name": "team-a-shared-bucket",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name =
'shared_logs'; ALLOW storage:logs:read WHERE storage:dt.security_context
MATCH ('team-a*');"
}

// Team B policy
{
  "name": "team-b-shared-bucket",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name =
'shared_logs'; ALLOW storage:logs:read WHERE storage:dt.security_context
MATCH ('team-b*');"
}
```

Pattern 3: Environment-Based Tiering

```
// Production access (restricted)
{
  "name": "production-access",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'prod_'; ALLOW storage:logs:read WHERE storage:dt.security_context
= 'env:production';"
}

// Non-production access (broader)
{
  "name": "non-production-access",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read WHERE
storage:dt.security_context MATCH ('env:dev*', 'env:staging*', 'env:qa*');"
}
```

Pattern 4: Multi-Dimensional MATCH() for Transversal Teams

When teams need access by component type (database, networking, OS) across multiple applications, encode multiple dimensions into the security context and use MATCH() to slice across application boundaries:

```
// Database team: all DB-layer components across all applications (Grail)
{
  "name": "db-team-grail-access",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read WHERE
storage:dt.security_context MATCH ('comp:db*');"
}

// Application team: all component layers for one application (Grail)
{
  "name": "easytrade-team-access",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read WHERE
storage:dt.security_context MATCH ('*/app:easytrade');"
}
```

> MATCH() supports `*` wildcards at any position and is **storage domain only**. For Classic entity access (hosts, services), use `startsWith` per component type. The context format is `comp:<component>/bu:<business-unit>/app:<application>` — place the transversal dimension first so `startsWith` cuts correctly. See **IAM-05: Boundary Design** for the full two-domain pattern.

Policy Boundaries

Policy boundaries decouple the "what" (policy) from the "where" (conditions), enabling reusable condition sets that can be applied across multiple policies.

What Are Boundaries?

Aspect	Description
Purpose	Bundle conditions for reuse across multiple policies
Scope	Record-level and resource-level restrictions
Relationship	Always used together with a policy — boundaries alone don't restrict anything
Limit	Maximum 10 restrictions per boundary

How Boundaries Work

While **policies** define *which features and data* users can access, **boundaries** define *where* users can access them:

```
Policy: ALLOW storage:logs:read, storage:metrics:read, storage:spans:read
Boundary: storage:k8s.namespace.name = "production"
          storage:dt.host_group.id STARTSWITH "prod-"
```

When assigned together, the user gets read access to logs, metrics, and spans — but only for production namespace data on production host groups.

Boundary Rules

Rule	Detail
No AND operator	Each line is one condition (implicitly AND-combined)
No logical operators	For complex logic, use policy templating
Reusable	Same boundary can be applied to multiple policies
Max 10 restrictions	Create additional boundaries if more are needed

Creating Boundaries

1. Go to **Account Management** > **Identity & access management** > **Policy management**
2. Select the **Boundaries** tab
3. Select **Create boundary**
4. Enter boundary name and conditions (one per line)
5. Save and assign to group policies

Example: Regional Boundary

```
# EU Region Boundary
storage:bucket-name STARTSWITH "eu_"
storage:azure.subscription = "eu-subscription-id"
```

This boundary can then be applied alongside any policy (logs access, metrics access, etc.) to restrict all data access to the EU region.

> **Tip:** Use boundaries when you have the same set of conditions (e.g., "production only" or "EU region only") that need to apply to multiple policies. This avoids duplicating conditions across every policy definition.

Enterprise Architecture Patterns

Tiered Access Model

Geographic/Regulatory Model

Region	Buckets	Context	Policy
EU	eu_*	region:eu	bucket-name STARTSWITH 'eu_' AND security_context MATCH ('region:eu*')
US	us_*	region:us	bucket-name STARTSWITH 'us_' AND security_context MATCH ('region:us*')

> **Tip:** Use policy boundaries to define regional restrictions once, then apply the same boundary to all team policies within that region.

Testing Permissions

Best Practices

Practice	Rationale
Start with least privilege	Grant minimum required access
Use groups, not individuals	Easier management, consistent access
Use policy boundaries for reuse	Avoid duplicating conditions across policies
Document all policies	Audit trail and governance
Test policies before deployment	Prevent access issues
Regular access reviews	Remove unnecessary permissions
Use MATCH for array fields	Required for correct evaluation
Combine bucket + record level	Defense in depth

Next Steps

Continue with the ORGNZ series:
- **ORGNZ-08**: Grail Segments

References

- Configure advanced permissions with security context
- Policy boundaries
- IAM policy reference

- Enhance data management with Grail

This notebook was AI-generated from Dynatrace documentation and enterprise best practices. It is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.

DQL: Access Distribution Audit

Measure coverage of key access control fields across your log data:

```
// Verify accessible data distribution across all key access control
dimensions
fetch logs, from:-1h
| summarize
    total = count(),
    withSecurityContext = countIf(isNotNull(dt.security_context)),
    withNamespace = countIf(isNotNull(k8s.namespace.name)),
    withHostGroup = countIf(isNotNull(dt.host_group.id))
```